

Supplementary Material

A. Calculation of the contribution of interstitials vs. vacancies

The following calculation demonstrated that the interstitial term on equation 16 of the main text can be considered of negligible influence respect the vacancy term, due to the difference of three orders of magnitude between them.

$$\frac{dV_{void}}{dt} = 4\pi R\Omega[D_v(C_v - C_v^V) - D_i C_i]$$

At $t=0.0001$, $C_v^V \approx 0$. For copper, we know that:

$$D_v = 3.9e^{-7} cm^2 s^{-1} [1]$$

$$C_v \approx 2e^{-4} cm^{-3} [2]$$

$$D_i = 3.19e^{-5} cm^2 s^{-1} [1]$$

$$C_i \approx 6e^{-10} cm^{-3} [2]$$

$$D_v C_v \approx 7.8e^{-11} \text{ and } D_i C_i \approx 1.9e^{-14}, \text{ therefore } D_v C_v \gg D_i C_i$$

B. Thermal diffusivity and stopping powers

First table shows the calculation of the stopping powers heavy ion/proton ratio for each of the elements that compose the GRCop-84 alloy. At the end of the table, a composition-averaged ratio is shown.

	Ei (eV)	7000000								
Incoming	Cu	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)	Ratio Stot (%)	
Target	Cu	29	63.546	31201.4356	0.00016772	1.3754E-16	7.4921E-11	7.4921E-11		
	Ei (eV)	3000000								
Incoming	H	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)		
Target	Cu	29	63.546	1045171.56	6.691E-06	8.0889E-21	7.9373E-12	7.9373E-12	10.5941765	
	Ei (eV)	7000000								
Incoming	Cu	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)		
Target	Cr	24	51.9961	34671.3284	0.00015246	1.1629E-16	7.4921E-11	7.4921E-11		
	Ei (eV)	3000000								
Incoming	H	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)		
Target	Cr	24	51.9961	1296343.87	5.4778E-06	6.8751E-21	7.9373E-12	7.9373E-12	10.5941795	
	Ei (eV)	7000000								
Incoming	Cu	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)		
Target	Nb	41	92.90638	25167.6954	0.00020364	1.8416E-16	7.4921E-11	7.4921E-11		
	Ei (eV)	3000000								
Incoming	H	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)		
Target	Nb	41	92.90638	703039.722	9.6648E-06	1.0745E-20	7.9373E-12	7.9373E-12	10.5941699	
88/8/4 averaged composition									10.5941765	

S 1. Calculation of the stopping powers heavy ion/proton ratio for each of the elements that compose the GRCop-84 alloy. At the end of the table, a composition-averaged ratio is shown. Excel file is accessible at the repository of this paper [3].

This second table shows the calculated drop in thermal diffusivity for Cu-ion irradiation and proton irradiation, and the calculated ratio.

Cu ions 3+ Low rate DPA		
Beam	SAWf	Th.diff
before	5.57E+08	6.77E-06
after	5.55E+08	6.60E-06
off/on	-0.40	-2.46
Protons at Low rate DPA		
Beam	SAWf	Th.diff
before	5.54E+08	6.59E-06
after	5.54E+08	6.60E-06
off/on	-0.01	0.22
		-11.155581

S 2. Calculated drop of thermal diffusivities from heavy ion irradiation (top values) and proton irradiation (bottom values). Thermal diffusivities drop ratio between heavy ion and protons is shown in the lower right corner. Excel file available in the repository of this paper [3].

Both tables are available in an Excel file on the repository for this paper [3].

References

- [1] A. B. Sivak, D. N. Demidov, and P. A. Sivak, “Diffusion Characteristics of Self-Point Defects in Copper: Molecular Dynamic Study,” *Phys. Atom. Nuclei*, vol. 85, no. 7, pp. 1245–1255, Dec. 2022, doi: 10.1134/S1063778822070171.
- [2] E. Martínez, F. Soisson, and M. Nastar, “Point defect evolution under irradiation: Finite size effects and spatio-temporal correlations,” *Journal of Nuclear Materials*, vol. 539, p. 152233, Oct. 2020, doi: 10.1016/j.jnucmat.2020.152233.
- [3] E. Botica-Artalejo, G. M. Wallace, and M. P. Short, “Repository for GRCop Beam Pulsing paper.” Github, Nov. 2025. doi: 10.5281/zenodo.18615596.